1、A software company developed a set of sales management system for a shopping mall. When analyzing and designing the system, the developers found that it was often necessary to traverse the commodity data and customer data in the system. Please use iterator mode to design the sales management system.The UML diagram of the pattern is as follows:

AbstractIterator

```
package experiment18_1;

interface AbstractIterator {
    void next();
    boolean isLast();
    void previous();
    boolean isFirst();
    Object getNextItem();
}
```

AbstractObjectList

```
package experiment18_1;

import java.util.List;

abstract class AbstractObjectList {
```

```java
    protected List<Object> objects;


    public void addObject(Object obj){
        objects.add(obj);
    };
    public void removeObject(Object obj){
        objects.remove(obj);
    };
    public List<Object> getObjects(){
        return objects;
    };
    public abstract AbstractIterator createIterator();
}
```

Client

```java
package experiment18_1;


import java.util.ArrayList;
import java.util.List;


public class Client {
    public static void main(String[] args) {
        List<Object> products = new ArrayList<Object>();
```

```java
        products.add("Product 1");
        products.add("Product 2");
        products.add("Product 3");
        products.add("Product 4");

        ProductList productList = new ProductList(products);
        AbstractIterator iterator =
productList.createIterator();

        System.out.println("Forward iteration:");
        while (!iterator.isLast()) {
            System.out.println(iterator.getNextItem());
            iterator.next();
        }


    }
}
```

ProductIterator

```java
package experiment18_1;

import java.util.List;
```

```java
class ProductIterator implements AbstractIterator{
    private ProductList productList;
    private List<Object> products;
    private int cursor1;//正向遍历
    private int cursor2;//反向遍历

    public ProductIterator(ProductList list){
        this.productList = list;
        this.products = list.getObjects();
        this.cursor2 = products.size() - 1;
    }

    @Override
    public void next() {
        if(cursor1 < products.size()){
            cursor1++;
        }
    }

    @Override
    public boolean isLast() {
```

```
        return cursor1 == products.size();
    }

    @Override
    public void previous() {
        if (cursor2 > -1) {
            cursor2--;
        }
    }

    @Override
    public boolean isFirst() {
        return cursor2 == 0;
    }

    @Override
    public Object getNextItem() {
        return products.get(cursor1);
    }
}
```

ProductList

```java
package experiment18_1;

import java.util.List;

class ProductList extends AbstractObjectList{
    public ProductList(List<Object> products){
        this.objects = products;
    }

    @Override
    public AbstractIterator createIterator() {
        return new ProductIterator(this);
    }
}
```

Output

```
D:\JDK\bin\java.exe "-javaagent:D:\idea\IntelliJ I
Forward iteration:
Product 1
Product 2
Product 3
Product 4

Process finished with exit code 0
```